Cognitive Robotics

Lab Report (Session 1 & 2)

23BRS1177

Steps to Connect to the Bot:

Connect to the robot:

```
$ ssh veerobot@192.168.0.[ID]


Accessing the robot's topics in local terminal:

1. Create your own workspace on local system.

    student@jazzy-jalisco:~$ mkdir -p ~/xyz_ws/src

    student@jazzy-jalisco:~$ cd ~/xyz_ws

    student@jazzy-jalisco:~/xyz_ws$ colcon build

2. In the robot's SSH terminal:

    veerobot@beetlebot-126:~$ source ~/lyra_ws/install/setup.bash

    veerobot@beetlebot-126:~$ export ROS_DOMAIN_ID=126 //replace 126 with your robot ID

3. In every new terminal that you open:

    student@jazzy-jalisco:~/xyz_ws$ source ~/xyz_ws/install/setup.bash

    student@jazzy-jalisco:~/xyz_ws$ export ROS_DOMAIN_ID=126 //replace 126 with your
robot ID


Launch and Arm the bot:

1. Run this command in the robot's terminal to launch:

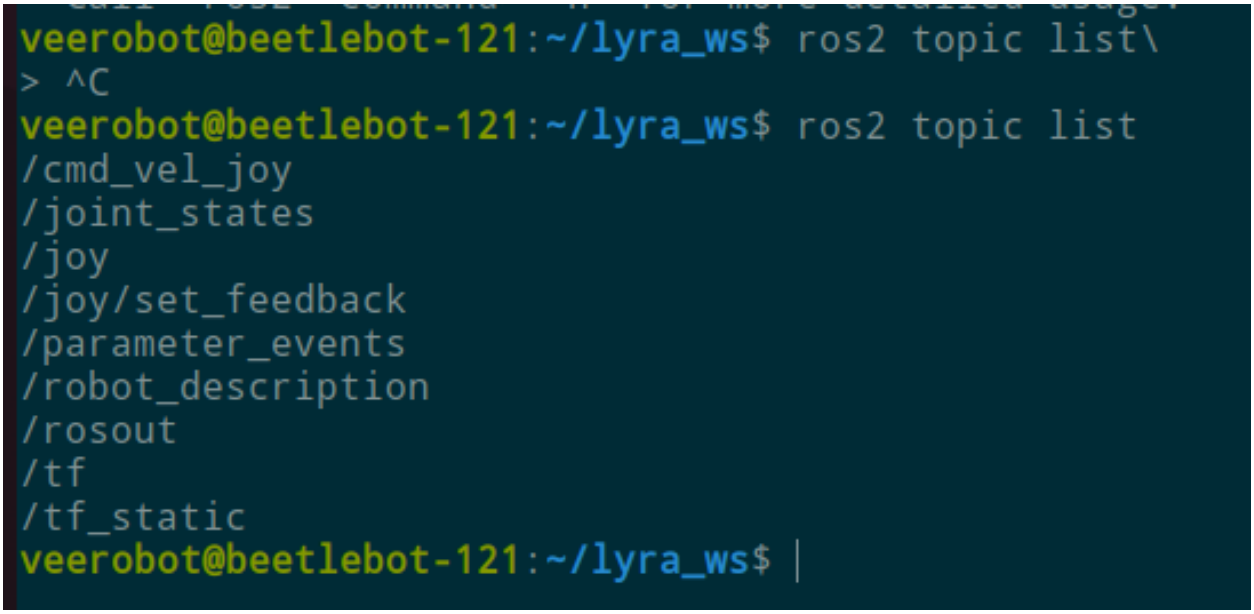
    veerobot@beetlebot-126:~$ ros2 launch lyra_bringup robot.launch.py mode:=slam
camera:=true imu:=true

2. Run this command in a sourced local terminal to arm:

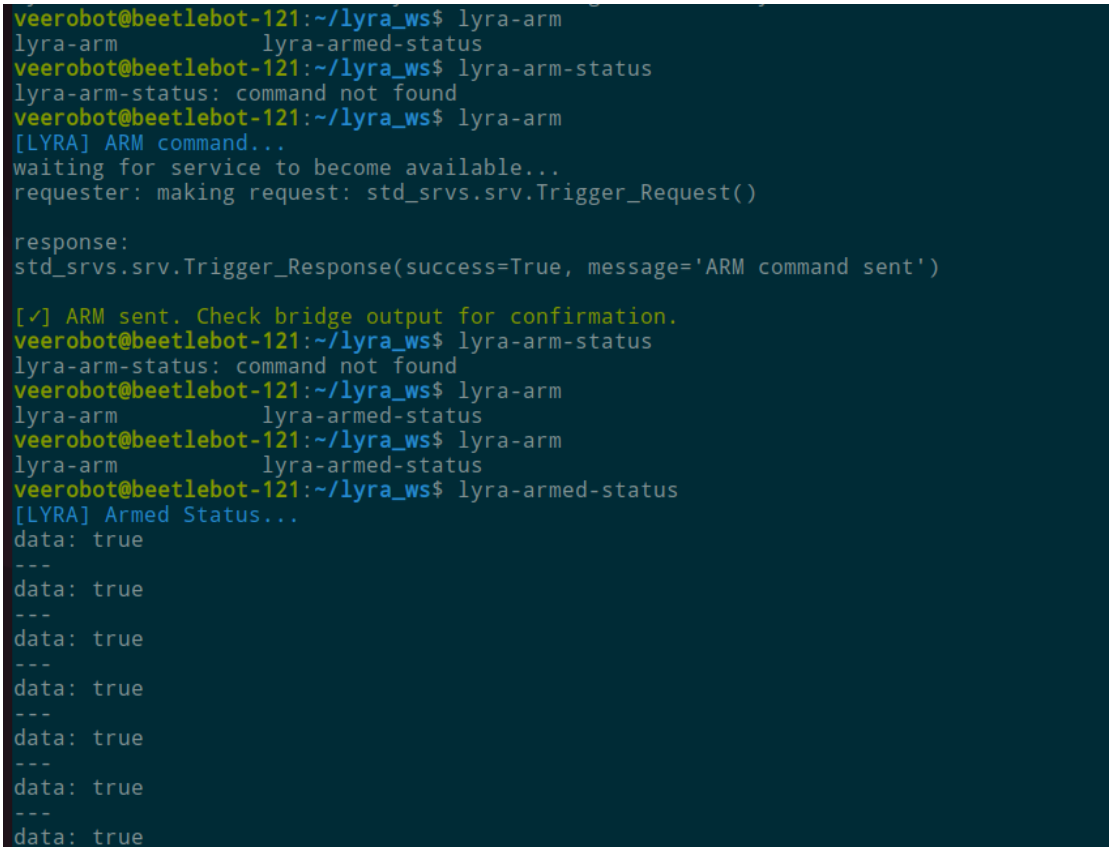
    student@jazzy-jalisco:~/xyz_ws$ ros2 service call /lyra/arm/std_srvs/srv/Trigger
```

Outputs:

List of Topics:



Arming The Robot:



Linear Movement and Teleoperation terminal screenshots:

```
veerobot@beetlebot-121:~/lyra_ws$   ros2 topic pub --once /cmd_vel_nav geometry_msgs/msg/Twist "{linear: {x: 0.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"
Waiting for at least 1 matching subscription(s)...
publisher: beginning loop
publishing #1: geometry_msgs.msg.Twist(linear=geometry_msgs.msg.Vector3(x=0.0, y=0.0, z=0.0), angular=geometry_msgs.msg.Vector3(x=0.0, y=0.0, z=0.0))

veerobot@beetlebot-121:~/lyra_ws$ ros2 run teleop_twist_keyboard teleop_twist_keyboard \
  --ros-args \
  -r cmd_vel:=/cmd_vel_nav

This node takes keypresses from the keyboard and publishes them
as Twist/TwistStamped messages. It works best with a US keyboard layout.
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Moving around:
   u    i    o
   j    k    l
   m    ,    .

For Holonomic mode (strafing), hold down the shift key:
---------------------------
   U    I    O
   J    K    L
   M    <    >

t : up (+z)
b : down (-z)

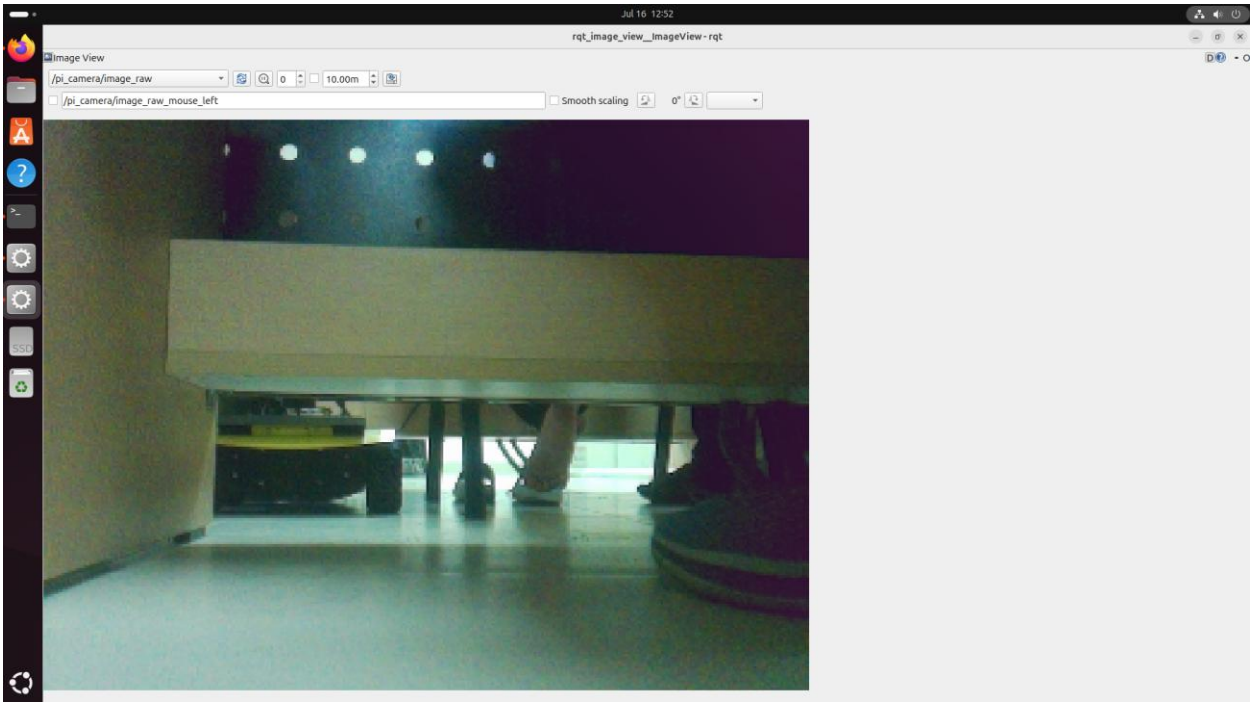
anything else : stop

q/z : increase/decrease max speeds by 10%
w/x : increase/decrease only linear speed by 10%
e/c : increase/decrease only angular speed by 10%

CTRL-C to quit

currently:      speed 0.50      turn 1.00
currently:      speed 0.45      turn 0.90
currently:      speed 0.41      turn 0.81
currently:      speed 0.36      turn 0.73
currently:      speed 0.33      turn 0.66
currently:      speed 0.30      turn 0.59
currently:      speed 0.27      turn 0.53
currently:      speed 0.24      turn 0.48
currently:      speed 0.22      turn 0.43
currently:      speed 0.19      turn 0.39
currently:      speed 0.17      turn 0.35
currently:      speed 0.16      turn 0.31
currently:      speed 0.14      turn 0.28
currently:      speed 0.13      turn 0.25
currently:      speed 0.11      turn 0.23
currently:      speed 0.10      turn 0.21
```

Camera Feed:



Videos on GitHub (Straight Line and Teleoperation):
https://github.com/Prayagraj27/Cognitive-Robotics-Lab